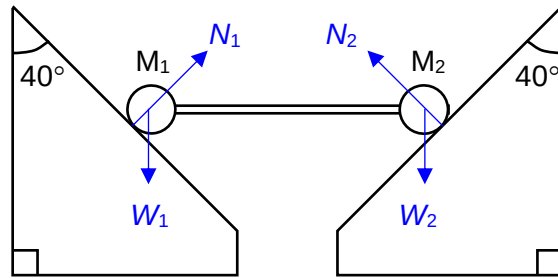


4 **(a)** Sum of the forces on the body in any direction must be zero. B1

Sum of the moments of the forces about any point on the body must be zero. B1

(b) (i)



correct identification of forces

B1

correct orientation of forces

B1

(ii) $N \cos 50^\circ = 2(9.81)$

M1

$N = 30.5 \text{ N}$

A1

(c) (i) The horizontal components of N_1 and N_2 must be equal in magnitude (so that horizontal net force is zero).

B1

(ii) Consider moments about the intersection of N_1 and N_3 .

B1

If bar is horizontal, clockwise moment produced by $4g$ is larger than the anti-clockwise moment produced by $2g$.

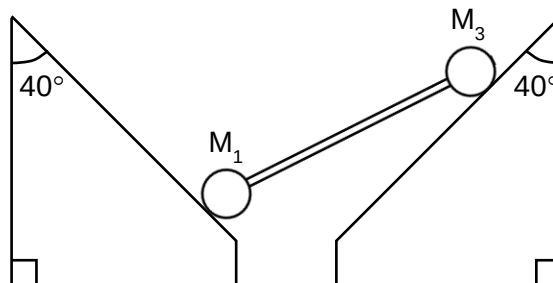
B1

OR

Consider the lines of action of N_1 , N_3 and $6g$.

They will not intersect at one point since the C.G. lies closer to M_3 than M_1 .

(iii)



[1